SOLUTION PROPOSAL · SYSTEM FLOW

After-Hours Telehealth & Home Doctor Platform

An end-to-end operating model for a Medicare bulk-billed after-hours medical service — covering the patient journey, the clinician workstation, dispatch and home visits, aged-care delivery, billing, and the regulatory integrations that make it lawful.

DOCUMENT	SCOPE	STATUS	VERSION
Overall System Flow	Platform & Delivery Plan	For Discussion	Draft 1.0

01 What the platform is

One sentence, so everything below has a frame.

The platform runs a **bulk-billed after-hours medical service**. It takes a patient who is unwell when their regular GP is closed, routes them to an available doctor within minutes, supports that doctor through a complete consultation — notes, prescribing, referral, investigation, certificates — and then closes the loop by delivering artefacts to the patient, a report to their usual GP, and a claim to Medicare.

It is **not** a video-calling product. Video is one channel inside a much larger clinical, logistical and financial workflow. The value sits in the routing, the clinical safety controls, the artefact pipeline and the billing engine.

CHANNEL 1

Telehealth

Phone or video consultation, delivered nationally. Highest volume, lowest cost to serve.

CHANNEL 2

Home Visits

A doctor physically attends. Geographically bounded, dispatch-routed, higher value per consult.

CHANNEL 3

Aged Care

Contracted cover for residential facilities. Booked by nursing staff through a facility portal, not by patients.

OPERATING WINDOW

The service exists specifically in the after-hours window — weeknights from 6pm to 8am, Saturdays from noon, and all day Sunday and public holidays. This is not a business preference. It is the window in which the after-hours Medicare item numbers apply, and it therefore defines when the platform generates revenue at all.

02 Who uses it

Six actors, five distinct interfaces. Most proposals underestimate this — the clinician console is only one of them.

ACTOR	INTERFACE	RESPONSIBILITY
Patient	Public site + booking app	Requests care, states symptoms and channel preference, provides Medicare details, receives artefacts.
Doctor	Clinician console	Claims patients from the queue, consults, prescribes, documents, bills. Contracted, not employed.
Dispatcher	Dispatch workstation	Phone triage, routes home-visit doctors, monitors queue health, escalates.
Facility staff	Aged-care portal	Requests after-hours cover for residents, supplies clinical context.
Clinical governance	Governance dashboard	Audits notes, tracks unactioned results, reviews prescribing outliers and incidents.
Administration	Admin console	Doctor credentialing, rostering, service-fee invoicing, business analytics.

Commercial model

Doctors are **independent contractors**. They bill Medicare under their own provider number; the operator takes a service fee collected by direct debit and invoiced weekly. The platform is therefore a two-sided marketplace with a clinical-safety obligation — it must attract and retain doctors while remaining accountable for the care they deliver.

03 The core operating flow

Nine stages from unwell patient to paid claim. Everything else in this document expands one of these.

01
Request
Patient books via web or phone. Symptoms, channel preference and Medicare details captured.

02
Triage
Eligibility validated, acuity scored, presenting complaint auto-categorised.

03
Queue
Patient enters the correct channel queue, ordered by acuity and wait time.

04
Claim
A doctor claims the patient. The record locks to them; family members assign together.

05
Connect
Phone or video established from inside the platform. Patient joins by SMS link.

06
Consult
History, examination, decision. Notes drafted by AI scribe against the doctor's template.

07
Act
Prescribe, refer, investigate, certify — each generating an artefact held in escrow.

08
Close
Doctor attests the notes and ends the consult. All artefacts release to the patient at once.

09
Settle
Claim batched to Medicare, report sent to the usual GP, service fee reconciled.

The single most important design rule

Nothing reaches the patient until the consult is ended. Prescriptions, referral letters, pathology forms and medical certificates all accumulate against the consult and release together on close. *End Consult* is a transaction boundary, not a screen change — it commits the clinical record, releases every artefact, and queues the claim. Modelling it any other way produces half-issued scripts and unbilled consults.

04 End-to-end journey by actor

The same flow, showing who does what and where the system acts on its own.

Patient

Books online or calls › Enters symptoms › Provides Medicare card › Consents to telehealth & recording › Receives SMS, joins consult › Consultation › Receives script QR, letters, certificate

Platform (automated)

Validates Medicare eligibility › Classifies presenting complaint › Scores acuity, ranks queue › Locks record on claim › Transcribes consult › Drafts notes to template › Releases artefacts on close › Batches claim, sends GP report

Doctor

Logs in, sets category filters › Reviews queue › Claims patient › Reads AI patient summary › Calls patient › Consults › Prescribes / refers / certifies › Reviews & attests AI notes › Selects item number, ends

Dispatcher

Takes phone bookings › Monitors queue depth & breaches › Routes home-visit doctors › Handles escalations from doctors › Responds to duress alerts

Governance & Admin

Credentials doctors › Audits notes & prescribing › Chases unactioned results › Reconciles service fees › Reports on demand & utilisation

05 Consult lifecycle

The state machine at the heart of the system. Every consult is in exactly one of these states.

Requested
Created at booking. Not yet eligible for the queue — awaiting eligibility and consent.

Queued
Visible to matching doctors. Acuity-ranked. Wait timer running against SLA.

Claimed
Locked to one doctor. Invisible to all others. Family members claim as a group.

In Consult
Connected by phone or video. Recording and transcription active. Duration timer feeding item eligibility.

Pending Attestation
Clinical activity complete; notes drafted but not yet signed. Cannot close from here.

Closed
Notes attested, artefacts released, claim queued, GP report generated. Record becomes append-only.

Requeued
Patient unreachable. Returns to the pool with attempt count incremented.

Rejected
Doctor declined with a recorded reason. Returns to queue; reason is auditable.

Abandoned
Exhausted contact attempts or patient withdrew. No claim. Retained for audit.

Attestation gate

A consult cannot move from *Pending Attestation* to *Closed* until the doctor has explicitly reviewed and signed the notes. Where AI drafted those notes, the system records the model version, retains the source transcript, and stores the doctor's sign-off. This is what makes an AI-assisted record defensible years later.

06 Inside the clinician console

Where a doctor spends an entire shift. Three zones, five actions.

ZONE 1

Clinical context

- Coded allergies
- AI patient summary across prior consults
- Conditions, long and short term
- Current medications
- Returned investigation results
- Personal note templates

ZONE 2

Documentation

- Template-driven note editor
- AI scribe from consult audio
- Per-doctor scribe personalisation
- Prior notes inline below
- Amendment history, append-only

ZONE 3

Patient channel

- Outbound phone from the platform
- Video once the patient joins
- Text chat and photo upload
- Nudge — rings the patient to join
- Add family, requeue, end

THE FIVE CLINICAL ACTIONS

ACTION	PRODUCES	DEPENDENCY
Prescribe	PBS or streamlined-authority eScript, delivered as a QR token	Drug database licence + eScript exchange + real-time prescription monitoring
Refer	Specialist or emergency-department referral letter	Secure messaging to the receiving provider
Investigate	Pathology or radiology request form	Results ingestion with acknowledgement tracking
Document	Medical certificate (work, school, carer, fit-to-return) or free-text instructions	Doctor's digital signature; verification endpoint
Bill	Medicare item selection, validated against consult time and duration	Claiming integration with rejection handling

07 Home visit & dispatch flow

The logistics half of the business. Same clinical core, entirely different operational shape.

01
Request
Patient books a visit or a telehealth doctor escalates one via dispatcher chat.

02
Geo-check
Address validated against the serviced coverage area for that city.

03
Dispatch
Job offered to doctors on the road; dispatcher can assign directly.

04
Route
Accepted jobs sequenced by route optimisation. Patient notified of ETA.

05
Attend
Doctor checks in on arrival. Duress timer armed. Consult conducted in person.

06
Close
Check-out, notes attested, artefacts released, next job served.

ADDITIONAL REQUIREMENTS

What home visits need that telehealth does not

- Mobile-first interface, used from a vehicle
- Offline tolerance — coverage is unreliable
- Live location and ETA to the patient
- Controlled-drug register for the doctor's bag
- Address safety flags

LONE-WORKER SAFETY

A duty of care, not a feature

Doctors attend unfamiliar addresses alone, at night. A reactive panic button requires a free hand and a conscious decision. The platform should arm a check-in timer on arrival and escalate automatically if the doctor does not check out — with the panic button as the backstop, not the primary control.

08 Aged care flow

Contracted B2B delivery. The requester is a nurse, not a patient — which changes the entire front end.

01

Facility requests

Nursing staff raise a request through the facility portal for a named resident.

02

Context supplied

Observations, medication chart, advance care directive and resuscitation status attached.

03

Routed

Into the nursing-home queue — visit or telehealth depending on availability and acuity.

04

Consult & chart

Doctor consults with staff present, charts medication, documents back to the facility record.

> **Why this is a separate build, not a queue tab**
>
> The consenting party is a facility, the clinical context arrives from a third party, the common workload is different (falls review, medication charting, palliative care, UTIs), and the output has to land in the facility's own records. Treating aged care as a filter on the patient queue will fail on all four counts.

09 Integration map

The platform is a hub. Most of its obligations are discharged through external systems.

Telehealth Platform Core

Patients · Consults · Clinical records · Artefacts · Claims · Audit

SYSTEM	DIRECTION	PURPOSE	STATUS
Medicare / claiming services	Two-way	Eligibility check, claim submission, rejection handling	BLOCKING
eScript exchange	Outbound	Electronic prescription tokens redeemable at any pharmacy	BLOCKING
Drug database	Inbound	Product data, dosing, interactions, contraindications	BLOCKING
Real-time prescription monitoring	Two-way	Monitored-medicine checks — legally mandated, per state	BLOCKING
Secure clinical messaging	Outbound	Consult report to the patient's usual GP; specialist referrals	CORE PROMISE
National health record	Two-way	Event summaries out, shared health summary in	EXPECTED
Pathology & radiology	Inbound	Results delivery into the ordering doctor's inbox	SAFETY-CRITICAL
Telephony & SMS	Two-way	Outbound calls, join links, artefact delivery, nudges	STANDARD
Video (WebRTC)	Two-way	Video consultation with recording for the scribe	STANDARD
Speech & language models	Outbound	Transcription, note drafting, summarisation, triage classification	STANDARD
Mapping & routing	Outbound	Coverage validation, route optimisation, live ETA	STANDARD
Interpreter service	Two-way	Three-way interpreted consultation	EQUITY
Payments & direct debit	Two-way	Private consult fees; doctor service-fee collection	STANDARD

10 Revenue flow

How a consultation becomes money, and where it can leak.

01

Item selected

Doctor picks an MBS item; the system validates it against consult time, duration and doctor type.

02

Claim queued

On consult close, the claim is written to the batch under the doctor's provider number.

03

Submitted

Daily batch to Medicare. Bulk-billed, so no patient payment.

04

Reconciled

Acceptances posted to the doctor's ledger. Rejections routed for correction and resubmission.

05

Fee applied

Operator service fee calculated on settled billings.

06

Collected

Weekly invoice issued and direct debit taken from the doctor.

Two leaks to close in the design

Unbilled consults. A consult closed without an item number is clinical work performed for nothing. The close flow must force an explicit item selection or an explicit no-billing reason.

Silent rejections. Batch-submit with no rejection workflow means failed claims are simply never paid and nobody notices. Rejection handling is revenue protection, not an edge case.

11 Clinical safety controls

The controls that must be designed in from the first sprint, because they cannot be retrofitted.

CONTROL	WHAT IT DOES	WHY IT CANNOT WAIT
Coded allergies	Structured allergy records rather than free text	Interaction checking is impossible against a text box; retrofitting means re-collecting every record
Interaction checking	Contraindication, duplicate-therapy and allergy alerts at the point of prescribing	The highest-consequence failure mode in the entire platform
Prescription monitoring	Checks monitored medicines against the state register	Legally mandated; prescribing without it is non-compliant
Paediatric dosing	Weight-based dose calculation with age guards	After-hours demand is heavily paediatric
Results acknowledgement	Tracks every ordered investigation to an acknowledged result, escalating when unactioned	Policy alone does not catch a missed abnormal result
Append-only records	Notes amendable only with visible, attributed history	A silently editable record has no evidentiary value
AI provenance	Records what the model produced, what the doctor changed, and the sign-off	Without it, no AI-drafted note is defensible in a complaint
Immutable audit log	Every access, edit, print and export attributed and timestamped	A privacy obligation and a breach-investigation prerequisite
Emergency escalation	One-click emergency pathway carrying patient location	Deteriorating patients present to after-hours services routinely

A structural risk worth naming early

When doctors can freely decline patients, permanently hide patients, filter by presenting complaint, and see their own earnings on every screen, the system quietly optimises against low-value patients. They sit in the queue while higher-value ones are claimed. The design answer is forced assignment above a wait threshold, dispatcher override, and visibility of decline rates to clinical governance.

12 System composition

Five deliverable surfaces over one clinical core.

SURFACE

Patient booking app

Request, triage questions, eligibility, consent, artefact retrieval, consult history.

SURFACE

Clinician console

Queue, consult workspace, clinical actions, inbox, history, billing, profile.

SURFACE

Dispatch workstation

Phone triage, allocation, doctor tracking, escalation and duress response.

SURFACE

Facility portal

Aged-care request intake, resident context, returned documentation.

SURFACE

Admin & governance

Credentialing, rostering, clinical audit, analytics, fee reconciliation.

CORE

Clinical platform

Identity, patients, consults, records, artefacts, claims, audit, integrations, real-time queue and messaging.

CROSS-CUTTING CONCERNS

CONCERN	REQUIREMENT
Real-time	Live queue, presence, chat and consult state pushed to all connected clients without polling
Concurrency	Record locking so two doctors cannot claim the same patient
Resilience	Mid-consult disconnection must restore state, never orphan a consult
Data residency	All health data stored onshore
Access control	Role-based, least privilege, with justified break-glass access
Authentication	Multi-factor mandatory for any account able to prescribe
Retention	Seven years for adults; until age twenty-five for minors
Accessibility	WCAG 2.1 AA across all patient-facing surfaces

13 Delivery approach

Sequenced so that regulatory dependencies run in parallel with build, not after it.

PHASE 1 **Clinical core**	Identity and role-based access; patient and consult data model; real-time waiting room with claim locking; consult workspace; templated notes with append-only history; manual billing capture. Audit logging and consent capture are built here, not later. Outcome: a doctor can run a complete consult and record it defensibly.
PHASE 2 **Communications**	Telephony, WebRTC video, SMS, patient join flow, nudge, clinical and dispatcher chat, notifications. Outcome: consults are conducted end-to-end inside the platform.
PHASE 3 **Clinical actions**	Certificates and documents first — no external dependency. Then investigations and referrals with secure messaging. Prescribing last, gated on the drug database, eScript exchange and prescription monitoring being contracted and conformance-tested. Outcome: the full artefact pipeline, released atomically on consult close.
PHASE 4 **Intelligence**	Transcription; AI scribe mapped to each doctor's template; cross-consult patient summary; triage classification and acuity scoring — every one shipped behind provenance capture and an attestation gate. Outcome: documentation time per consult materially reduced.
PHASE 5 **Scale surfaces**	Patient booking app; dispatch workstation with routing and lone-worker safety; aged-care facility portal; governance and analytics. Outcome: all three revenue channels operating, with oversight.

Regulatory dependencies — start immediately

These carry lead times measured in months and involve commercial agreements and conformance testing. They are the critical path for Phase 3, and they should be initiated on day one of Phase 1.

- Approved deputising-service status, without which the after-hours item numbers are not billable
- Drug database licensing
- Electronic prescription exchange conformance
- Medicare online claiming registration and device activation
- Real-time prescription monitoring access, obtained per state
- National health record conformance assessment
- Secure clinical messaging vendor agreement

> **Recommended first milestone**
>
> A working Phase 1 vertical slice — one doctor, one queue, one consult, fully recorded and audited — is the fastest way to validate the model with real clinicians before committing to the integration spend. Everything after that is addition, not rework.

14 Principal risks

RISK	SEVERITY	MITIGATION
Prescribing without interaction and allergy checking	CRITICAL	Coded allergies from the first schema; interaction checking shipped with the prescribing module, never after it
Ordered investigations never followed up	CRITICAL	Acknowledgement tracking with automatic escalation to clinical governance
AI-drafted notes disputed later	CRITICAL	Transcript retention, model versioning, mandatory attestation before close
Regulatory approvals delay launch	CRITICAL	Applications initiated in parallel with Phase 1; prescribing deliberately sequenced last
Low-acuity patients stranded in queue	HIGH	Acuity ranking, forced assignment above a wait threshold, decline-rate visibility
Claims rejected and never resubmitted	HIGH	Rejection queue with correction workflow and ageing report
Lone-worker incident on a home visit	HIGH	Arrival check-in with automatic escalation on missed check-out; panic button as backstop
Doctor supply insufficient at peak	HIGH	Demand forecasting, rostering, and surge incentives surfaced in the admin console
Health data breach	HIGH	Onshore residency, encryption, least-privilege access, immutable audit, tested breach response